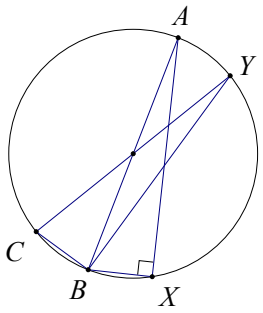
1 Team Problems

Problem 1. Compute the number of ordered triples of positive integers (a, b, c) such that $\sqrt{a^b + c!} = 28$.

Problem 2. Chris the frog begins on a number line at 0. Chris takes jumps of lengths $1, 2, 3, \ldots, 2017$, in that order. If Chris's current location is an even integer, he jumps in the positive direction; otherwise, he jumps in the negative direction. Let $P(n)$ denote Chris's location after the nth jump. Compute $\sum_{j=1}^{2017} P(j)$.

Problem 3. The diagram below shows arc $\overset{\frown}{ABC}$, which has a measure of 210°. Points X and Y lie on the arc so that $\text{m}\angle AXB = 90°$ and $\triangle ABX \cong \triangle YCB$. Given that $AX = 8$, the value of $[ABC]$ can be expressed in the form $a + b\sqrt{c}$, where a, b, c are integers and c is not divisible by any perfect square greater than 1. Compute the ordered triple (a, b, c).



Problem 4. Ari repeatedly rolls a standard, fair, six-sided die. Let $R(n)$ be the nth number rolled, and let $Q(n) = R(1) \cdot R(2) \cdot \ldots \cdot R(n)$. Compute the probability that there exists an n such that $Q(n) = 100$ and for all $m < n$, $Q(m)$ is not a perfect square.

Problem 5. Given that C, A, T, F, I, S, and H are digits, not necessarily distinct, and that

$$3 \cdot \underline{C}\,\underline{A}\,\underline{T} \cdot \underline{F}\,\underline{I}\,\underline{S}\,\underline{H} = \underline{C}\,\underline{A}\,\underline{T}\,\underline{F}\,\underline{I}\,\underline{S}\,\underline{H},$$

compute the greatest possible value of $\underline{C}\,\underline{A}\,\underline{T}\,\underline{F}\,\underline{I}\,\underline{S}\,\underline{H}$.

Problem 6. Let $\{a_n\}$ be a sequence with $a_0 = 1$, and for all $n > 0$, $a_n = \frac{1}{2}\sum_{i=0}^{n-1} a_i$. Compute the greatest value of n for which $a_n < 2017$.

Problem 7. On July 17, 2017, the nation of Armlandia will turn n^2 years old and the nation of Nysmlistan will turn n years old. The next four anniversaries for which Nysmlistan's age divides Armlandia's age will occur, in order, on July 17 in the years 2027, 2032, 2038, and M. Compute M.

Problem 8. Ellipse $\mathcal{E}$ has center O, major axis of length 10, and minor axis of length 4. Ellipse $\mathcal{E}'$ is obtained by rotating $\mathcal{E}$ counterclockwise about O by 60°. Proceeding clockwise around the perimeter of $\mathcal{E}$, the intersection points of $\mathcal{E}$ and $\mathcal{E}'$ are labeled A, R, M, L. Compute $[AOL]$.

Problem 9. Let $E(n)$ denote the least integer strictly greater than n whose base-10 representation contains only even digits, and let $O(n)$ denote the least integer strictly greater than n whose base-10 representation contains only odd digits. Compute the least positive integer N for which $E(N) + O(N)$ is a multiple of 2017.

Problem 10. Compute the number of tilings of a 4×7 rectangle using only 1×1, 2×2, 3×3, and 4×4 tiles.